

SHAMIR KHAN

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Education

Bachelor of Science in Computer Science

April 2028

Wilfrid Laurier University

Waterloo, ON

- **Core Coursework:** Data Structures, Algorithms, OOP, Discrete Mathematics, Computer Architecture, Digital Electronics

Technical Experience

Full-Stack Developer — Team Lead

April 2026

ConHacks | InstantService

Waterloo, ON

- **Led a team of 4 developers** to architect an AI-native dispatch engine using **Gemini 1.5 Pro**, reducing client service-acquisition time by **70%** by automating natural language request classification and real-time contractor dispatch.
- **Engineered a multi-weighted contractor matching algorithm** with **FastAPI** and **Snowflake**, optimizing contractor allocation to drive a **40% increase in job volume** and deliver **instant, under-3-minute service confirmations**.
- **Architected a secure backend infrastructure** for tier-based pricing and contractor verification, automating verification of **license/insurance certificates** to establish a high-trust production environment.

Operations Logistics

Sept 2021 – Sept 2022

Hack the North (Canada's Largest Hackathon)

Waterloo, ON

- **Facilitated a seamless 36-hour event experience** for **1,000+** global students by managing on-site logistics, registration desks, and wayfinding operations across **5+** venue zones.
- **Orchestrated on-schedule attendee registration** and technical demo scheduling for **827** attendees across **222** concurrent projects, resolving critical bottlenecks with **zero major schedule disruptions**.

High-Impact Projects

CloudGuard | Cloud Security Posture Management (CSPM) Platform | *Live Site* | *GitHub*

July 2026

- **Architected a production-grade, 6-language polyglot microservices platform** (C++, Java, Python, Scala, Kotlin, TypeScript) deployed across **Render** and **Vercel**, achieving **99.9% uptime** with a persistent **Supabase PostgreSQL** backend.
- **Built a native C++ hardware monitoring agent** using **Windows APIs** (GlobalMemoryStatusEx, GetSystemTimes) and Linux **/proc** parsers, scraping real-time CPU/RAM utilization and publishing to **Apache Kafka** topics at **5-second intervals**.
- **Developed a Python IsolationForest ML anomaly detection pipeline** that processes Kafka streams to flag billing spikes and SSH brute-force intrusions, reducing mean-time-to-detect security threats by **85%**.
- **Engineered a Scala Spark streaming analytics engine** to aggregate real-time cost data across **3 cloud providers** (AWS, Azure, GCP) into sliding time windows, enabling sub-second billing anomaly correlation.

InstantService | AI-Powered Dispatch PWA | *Live Site* | *GitHub*

April 2026

- **Architected a mobile-first PWA and AI dispatch engine** using **Gemini 1.5 Pro** that **reduced client service-acquisition time by 70%** via automated natural language request classification and instant dispatch.
- **Engineered a multi-weighted contractor matching algorithm** with **FastAPI** and **Snowflake**, optimizing provider matching to drive a **40% increase in daily job volume** and deliver **under-3-minute service confirmations**.
- **Integrated Google Maps APIs** and live **WebSocket** feeds to calculate optimal service routing, reducing map rendering API overhead by **20%** with sub-second recalculation latency.

MSA Website | Digital Engagement Platform | *Live Site*

Jan 2026

- **Improved navigation flows and content accessibility** for a **250+ member community** platform, consulting on UI/UX product design for the official **React/Next.js** digital ecosystem.
- **Developed interactive event registration and fundraising modules** using **TypeScript** and **Payload CMS**, supporting community-driven initiatives that generated **\$50K+** in charitable donations.

Technical Skills Summary

Languages: Python, Java, C++, C, Kotlin, Scala, JavaScript, TypeScript, HTML, CSS, SQL

Frameworks & Libraries: React.js, Next.js, Spring Boot, FastAPI, Node.js, Express, Apache Spark, Tailwind CSS

AI & Data: Gemini 1.5 Pro, IsolationForest ML, ElevenLabs, Apache Kafka, Snowflake, MongoDB, PostgreSQL

Cloud & DevOps: Docker, Terraform, Vercel, Render, Supabase, Git, GitHub Actions, CI/CD Pipelines

Concepts: Microservices Architecture, Event-Driven Systems, Anomaly Detection, Stream Processing, REST APIs, JWT Auth